Oliver Barta

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Technical Skills

Languages: Python, C++, Java, R, HTML, CSS, Javascript
Designing: AutoCAD, SolidWorks, Z-Suite
Frameworks/Libraries: React, OpenCV, PyTorch, TensorFlow, Matplotlib, NumPy, Napari, Swing, Pandas, SciPy
Developer Tools: Git, CI/CD, VS Code

Work

Research Assistant — June 2025 - August 2025
Schlegel-UW Research Institute for Aging — *Waterloo, ON*

- **Frameworks/Libraries:** TensorFlow, NumPy, OpenCV, Matplotlib, Napari, SciPy, and MagicGUI
- **Developed an AI** that segments ultrasound footage of arteries, trained on 50000+ frames of data
- The Lab now uses the AI to segment hundreds of ultrasound videos for analyses
- Organized study participant data into 5 dimension graphs and matrices using R and Matlab code to present to other researchers

Skills Development Instructor — June 2023 - August 2024
Waterloo Minor Soccer Club — *Waterloo, ON*

- Worked in a high-paced, competitive coaching environment while playing soccer at a competitive level
- Enhanced leadership skills by mentoring soccer players

Projects

UFC Fighter Website | *HTML, CSS, Javascript, Python* — October 2025 – Present

- Built **a fully responsive UFC stats web app** using HTML, CSS, and JavaScript with live fighter search, animated UI transitions, and JSON data integration

Seat Planner | *Java, Swing, AWT, ImageIO, File I/O, CSV Parsing, Regex* — May 2025 – June 2025

- Designed and implemented **a Java Swing GUI application that automates creation of school assembly seating plans** from CSV datasets
- Developed a parsing engine to handle inconsistent teacher/class CSV input using regex and validation dialogs
- Utilized Java AWT Graphics2D and ImageIO to programmatically render class layouts onto auditorium blueprints

Banking System | *Java, Swing, AWT, CSV Parsing* — January 2025 – February 2025

- Developed an **interactive desktop banking system** in Java that loads financial data from a CSV file and allows users to search for account balances by name using linear or binary search
- Implemented the sorting algorithms: Bubble, Merge, Insertion, Quick, and Radix to sort datasets by various financial attributes
- Built a graphical user interface using Swing and AWT, including buttons, dropdowns, and input fields

Battleship | *Python, Pygame, File I/O, ImageIO* — April 2024 – June 2024

- Designed and programmed a fully interactive Battleship game using Pygame
- Developed opponent logic with randomized targeting and **hit-detection algorithms**

Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Systems Design Engineering, Faculty of Engineering — *September 2025 – Present*

- SYDE 121 - Digital Computation: Developed projects in **C++**
- SYDE 101L - Communications, Visualization: Design models in **SolidWorks** and **AutoCAD**
  - 3D-Printing models using Z-Suite
- Formula Electric Design team: Developed a driver in **C++** in **STM32Cube** for a digital to audio converter